## SECTION A – 2 MARKS

### Q1. (a)

**Statistics** is the science of collecting, presenting, analysing and interpreting numerical data. In the words of Croxton and Cowden, statistics may be defined as "the collection, presentation, analysis and interpretation of numerical data".

### Q1. (b)

1. **Primary data:** Data collected for the first time by the investigator himself, for a specific purpose. They are original in nature.
2. **Secondary data:** Data already collected by someone else and used by the investigator, such as published reports and records.

### Q1. (c)

**Tabulation** is the systematic arrangement of classified data in rows and columns. It presents a large mass of data in a compact and orderly form, so that it is easy to understand and compare.

### Q1. (d)

1. The sum of the deviations of the items from the arithmetic mean is always zero: $\Sigma(x - \bar{x}) = 0$.
2. The sum of the squares of the deviations from the arithmetic mean is the least: $\Sigma(x - \bar{x})^2$ is a minimum.

### Q1. (e)

Largest value L = 100, smallest value S = 20.
Range = L − S = 100 − 20 = **80**.
Coefficient of range = (L − S) / (L + S) = 80 / 120 = **0.67**.

### Q1. (f)

**Standard deviation** is the square root of the arithmetic mean of the squared deviations of the items from their arithmetic mean. It is denoted by σ: $\sigma = \sqrt{(\Sigma(x - \bar{x})^2 / N)}$. It is the most widely used measure of dispersion.

### Q1. (g)

**Correlation** is the relationship between two or more variables, such that a change in one variable is accompanied by a change in the other. For example, price and demand, or income and expenditure, are correlated.

### Q1. (h)

**Regression** is the statistical method of estimating the unknown value of one variable from the known value of another variable. It shows the average relationship between two variables in the form of an equation.

### Q1. (i)

**Classification** is the process of arranging the collected data into groups or classes according to their common characteristics, such as area, time, attribute or size.

### Q1. (j)

A **histogram** is a graph of a frequency distribution in which the class intervals are shown on the X-axis and the frequencies on the Y-axis. Each class is shown by a rectangle whose height is its frequency, and the rectangles are drawn next to one another.

## SECTION B – 5 MARKS

### Q2.

1. **Geographical classification:** Data are grouped by place or area, such as country, state, district or city, for example the population of each state.
2. **Chronological classification:** Data are grouped by time, such as years, months or weeks, for example the sales of a company year by year.
3. **Qualitative classification:** Data are grouped by an attribute that cannot be measured, such as sex, religion, literacy or occupation.
4. **Simple and manifold classification:** In simple classification data are divided by one attribute into two classes (for example literate and illiterate); in manifold classification they are divided by two or more attributes at a time.
5. **Quantitative classification:** Data are grouped by a characteristic that can be measured, such as height, weight, income or marks, using class intervals. Such a grouping is a frequency distribution.

### Q3.

1. **Direct personal investigation:** The investigator collects the information personally from the persons concerned. It suits a small, local enquiry and gives accurate data.
2. **Indirect oral investigation:** Information is collected from third persons (witnesses) who know the facts, as in enquiries by committees and commissions.
3. **Information through local correspondents:** Local agents or correspondents are appointed in different areas, and they send the information regularly, as newspapers do.
4. **Mailed questionnaire:** A list of questions is sent by post to the informants, who fill it in and return it. It covers a wide area at low cost.
5. **Schedules sent through enumerators:** Trained enumerators take the schedule to the informants, ask the questions and fill in the answers themselves, as in the census.

### Q4.

| x | f | cf |
|---|---|---|
| **20** | 25 | 25 |
| **60** | 48 | 73 |
| **80** | 60 | 133 |
| **100** | 57 | 190 |
| **120** | 29 | 219 |
| **140** | 10 | 229 |
| **160** | 8 | 237 |
| **200** | 2 | 239 |
| **Total** | **N = 239** | |

Q1 = size of (N + 1)/4 th item = 240/4 = 60th item. The 60th item lies in cf 73, so **Q1 = 60**.
Q3 = size of 3(N + 1)/4 th item = 3 × 60 = 180th item. The 180th item lies in cf 190, so **Q3 = 100**.
Quartile deviation = (Q3 − Q1) / 2 = (100 − 60) / 2 = **20**.
Coefficient of QD = (Q3 − Q1) / (Q3 + Q1) = 40 / 160 = **0.25**.

### Q5.

| x | log x |
|---|---|
| **20** | 1.3010 |
| **350** | 2.5441 |
| **450** | 2.6532 |
| **600** | 2.7782 |
| **861** | 2.9350 |
| **958** | 2.9814 |
| **N = 6** | **Σ log x = 15.1929** |

GM = Antilog (Σ log x / N) = Antilog (15.1929 / 6) = Antilog 2.5322
**GM = 340.5**

### Q6.

**Title:** Distribution of employees in an office according to sex, salary grade and designation

| Designation | Sex | Below 3,000 | 3,000 to 5,000 | 5,000 and above | Total |
|---|---|---|---|---|---|
| **Supervisor** | Male | | | | |
| | Female | | | | |
| | Total | | | | |
| **Assistant** | Male | | | | |
| | Female | | | | |
| | Total | | | | |
| **Clerk** | Male | | | | |
| | Female | | | | |
| | Total | | | | |
| **Grand total** | | | | | |

The column headings show the salary grade (Rs.), the stub shows designation and sex, and each cell will hold the number of employees.

**Source:** ..... **Note:** .....

### Q7.

N = 20, $\bar{x} = 4$, $\bar{y} = 2$, $\Sigma x^2 = 1{,}680$, $\Sigma y^2 = 320$, $\Sigma xy = 480$.

1. **Variance of x:** $\sigma x^2 = \Sigma x^2/N - \bar{x}^2 = 1{,}680/20 - 4^2 = 84 - 16 = 68$.
2. **Variance of y:** $\sigma y^2 = \Sigma y^2/N - \bar{y}^2 = 320/20 - 2^2 = 16 - 4 = 12$.
3. **Covariance:** $\Sigma xy/N - \bar{x}\bar{y} = 480/20 - (4 \times 2) = 24 - 8 = 16$.
4. **Regression coefficients:** $b_{xy} = 16/12 = \mathbf{1.333}$; $b_{yx} = 16/68 = \mathbf{0.235}$.

**Regression equation of x on y:** $x - \bar{x} = b_{xy}(y - \bar{y})$
$x - 4 = 1.333(y - 2)$, so $\mathbf{x = 1.333y + 1.333}$

**Regression equation of y on x:** $y - \bar{y} = b_{yx}(x - \bar{x})$
$y - 2 = 0.235(x - 4)$, so $\mathbf{y = 0.235x + 1.06}$

**When y = 7:** $x = 1.333 \times 7 + 1.333 = \mathbf{10.67}$

## SECTION C – 10 MARKS

### Q10.

The classes are inclusive, so they are changed to exclusive classes (9.5 – 19.5 and so on) for the median and the mode.

| Class | Exclusive class | f | Mid-point (x) | fx | cf |
|---|---|---|---|---|---|
| **10 – 19** | 9.5 – 19.5 | 10 | 14.5 | 145.0 | 10 |
| **20 – 29** | 19.5 – 29.5 | 18 | 24.5 | 441.0 | 28 |
| **30 – 39** | 29.5 – 39.5 | 8 | 34.5 | 276.0 | 36 |
| **40 – 49** | 39.5 – 49.5 | 6 | 44.5 | 267.0 | 42 |
| **50 – 59** | 49.5 – 59.5 | 4 | 54.5 | 218.0 | 46 |
| **60 – 69** | 59.5 – 69.5 | 2 | 64.5 | 129.0 | 48 |
| **70 – 79** | 69.5 – 79.5 | 5 | 74.5 | 372.5 | 53 |
| **80 – 89** | 79.5 – 89.5 | 7 | 84.5 | 591.5 | 60 |
| **Total** | | **N = 60** | | **Σfx = 2,440** | |

**Mean**

$\bar{x} = \Sigma fx / N = 2{,}440 / 60 = \mathbf{40.67}$

**Median**

$N/2 = 60/2 = 30$th item, which lies in the class 29.5 – 39.5 (cf 36).

$L = 29.5$, $cf = 28$, $f = 8$, $i = 10$

$\text{Median} = L + ((N/2 - cf)/f) \times i = 29.5 + ((30 - 28)/8) \times 10 = 29.5 + 2.5 = \mathbf{32}$

**Mode**

The highest frequency is 18, so the modal class is 19.5 – 29.5.

$L = 19.5$, $f_1 = 18$, $f_0 = 10$, $f_2 = 8$, $i = 10$

$\text{Mode} = L + ((f_1 - f_0)/(2f_1 - f_0 - f_2)) \times i = 19.5 + (8/(36 - 10 - 8)) \times 10 = 19.5 + (8/18) \times 10 = 19.5 + 4.44 = \mathbf{23.94}$

**Mean = 40.67, Median = 32, Mode = 23.94**

### Q11.

Assumed mean A = 35, class interval i = 10, $d = (m - 35)/10$.

| Class | f | Mid-point (m) | d | fd | $fd^2$ |
|---|---|---|---|---|---|
| **10 – 20** | 10 | 15 | −2 | −20 | 40 |
| **20 – 30** | 18 | 25 | −1 | −18 | 18 |
| **30 – 40** | 20 | 35 | 0 | 0 | 0 |
| **40 – 50** | 6 | 45 | 1 | 6 | 6 |
| **50 – 60** | 4 | 55 | 2 | 8 | 16 |
| **60 – 70** | 2 | 65 | 3 | 6 | 18 |
| **Total** | **N = 60** | | | **Σfd = −18** | **$\Sigma fd^2 = 98$** |

**Mean**

$\bar{x} = A + (\Sigma fd / N) \times i = 35 + (-18/60) \times 10 = 35 - 3 = \mathbf{32}$

**Standard deviation**

$\sigma = \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i = \sqrt{(98/60 - (-18/60)^2)} \times 10$
$\sigma = \sqrt{(1.6333 - 0.09)} \times 10 = \sqrt{1.5433} \times 10 = 1.2423 \times 10 = \mathbf{12.42}$

**Coefficient of variation**

$CV = (\sigma / \bar{x}) \times 100 = (12.42/32) \times 100 = \mathbf{38.82\%}$

### Q12.

| x | y | $x^2$ | $y^2$ | xy |
|---|---|---|---|---|
| **8** | 25 | 64 | 625 | 200 |
| **10** | 30 | 100 | 900 | 300 |
| **15** | 32 | 225 | 1,024 | 480 |
| **17** | 35 | 289 | 1,225 | 595 |
| **20** | 37 | 400 | 1,369 | 740 |
| **22** | 40 | 484 | 1,600 | 880 |
| **24** | 42 | 576 | 1,764 | 1,008 |
| **25** | 45 | 625 | 2,025 | 1,125 |
| **$\Sigma x = 141$** | **$\Sigma y = 286$** | **$\Sigma x^2 = 2{,}763$** | **$\Sigma y^2 = 10{,}532$** | **$\Sigma xy = 5{,}328$** |

N = 8

**Coefficient of correlation**

$r = (N\Sigma xy - \Sigma x\Sigma y) / \sqrt{[(N\Sigma x^2 - (\Sigma x)^2)(N\Sigma y^2 - (\Sigma y)^2)]}$
$N\Sigma xy - \Sigma x\Sigma y = 8 \times 5{,}328 - 141 \times 286 = 42{,}624 - 40{,}326 = 2{,}298$
$N\Sigma x^2 - (\Sigma x)^2 = 8 \times 2{,}763 - 141^2 = 22{,}104 - 19{,}881 = 2{,}223$
$N\Sigma y^2 - (\Sigma y)^2 = 8 \times 10{,}532 - 286^2 = 84{,}256 - 81{,}796 = 2{,}460$
$r = 2{,}298 / \sqrt{(2{,}223 \times 2{,}460)} = 2{,}298 / \sqrt{54{,}68{,}580} = 2{,}298 / 2{,}338.5 = \mathbf{0.983}$

**Probable error**

$PE = 0.6745 \times (1 - r^2)/\sqrt{N} = 0.6745 \times (1 - 0.9657)/\sqrt{8} = 0.6745 \times 0.0343/2.828 = \mathbf{0.0082}$

**r = 0.983, PE = 0.0082.** There is a very high positive correlation between x and y. Since r is more than 6 times the PE ($6 \times 0.0082 = 0.049$), the correlation is significant.